1 POTENTIOMETER AS ANGLE SENSOR

1.1 Tasks:

1. Add the potentiometer to the shaft of barrier gate. You can follow the instrucions in the video.

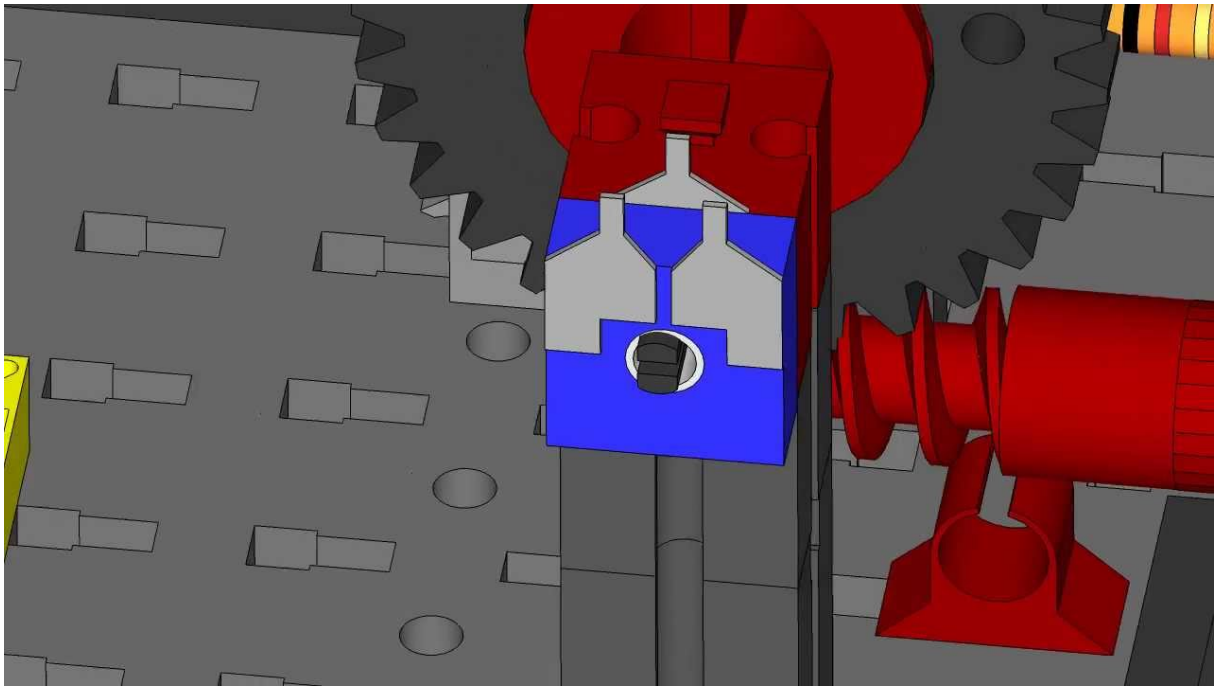

Figure 1: Adding potenciometer as an angle sensor.

2. Test the potentiometer values with next program:

```
void setup() {
 Serial.begin(9600);
}

void loop() {
 Serial.println(analogRead(A3));
 delay(100);
}
```

3. Change the functions for lifting and lowering the barrier gate to use potenciometer readings instead of switch and time controlled movement.

```
    [+] void setup() {
    [+] void loop() {
    [+] void manualGateControll(){
    [+] void stopTheGate(){
    [-] void moveGateUp() {
            int gate_orientation = analogRead(POTENTIOMETER_PIN);
            while (gate_orientation < 750){
                digitalWrite(MOTOR_PIN_1, HIGH);
                digitalWrite(MOTOR_PIN_2, LOW);
                gate_orientation = analogRead(POTENTIOMETER_PIN);
            }
            stopTheGate();
        }
    [+] void moveGateDown() {
```

4. Advanced: Calculate the angle of barrier gate from the analog readings of potenciometer.

1.2 Questions:

1. What is the value of the angle sensor when the barrier gate is in the upper orientation...
2. ... and in lower orientation.

1.3 Summary:

1.3.1 <++>

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1.4 Issues:

1.4.1 <++>

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